14/2/2026

Embeddings :

Food - recommendation app

Movie recommendation app *

Restaurant recommendation

25000 Items 500,000

PI SH HO SA SO

pizza, shawarma, hotdog, salad, soup

[1 0 0 0 0]

SOUP	PIZZA	SH	HO	SA

SOUP → vector → [1, 0, 0, 0, 0]

↓

one hot encoded representation

Hotdog	0 0 0 1 0

Issues / Problems

① Number of weights

M entries one hot encoding

N nodes in the first layer

$M \times N$ ✓

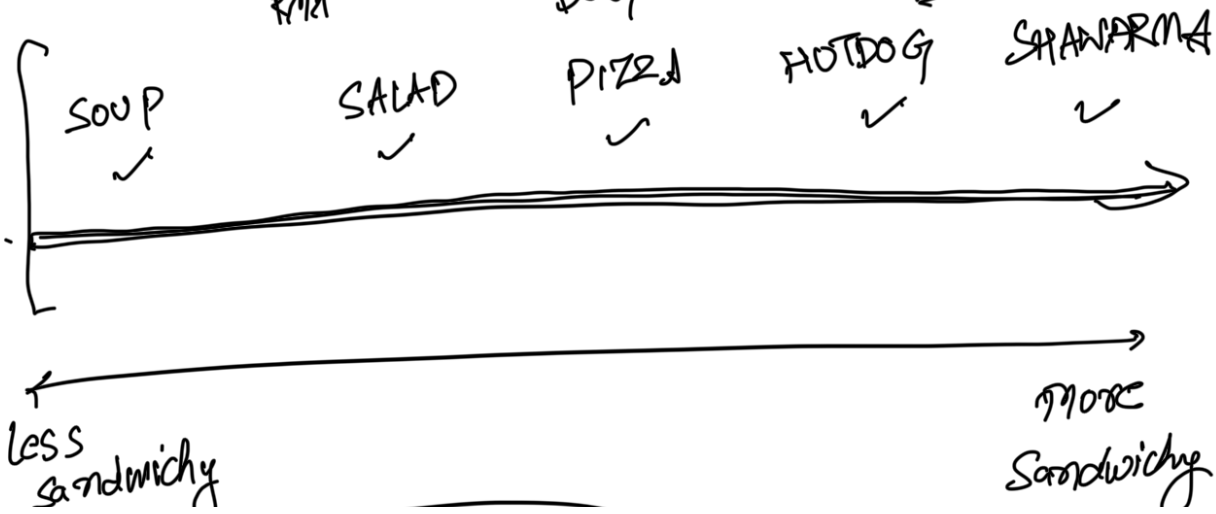
2. Number of dataset / points will also increase
3. Number of computations will also increase
4. Memory.
5. Nvidia-Jetson / on-device ML

Embedding: An embedding is a vector representation of data in embedding space.

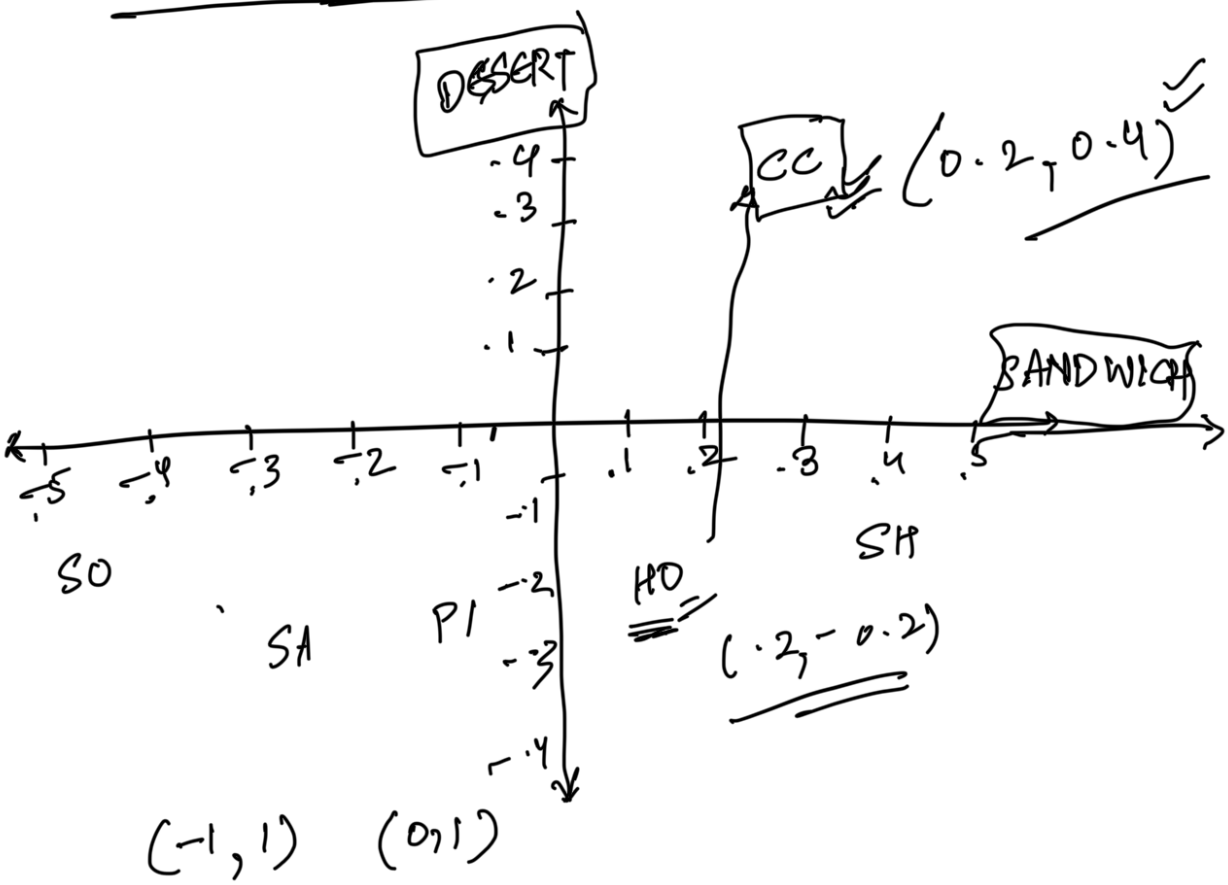
* Embeddings make it easy to do machine learning on large feature vectors
  - ↳ MOVIE VECTORS
  - ↳ FOOD VECTORS

PI SH SO HO SA

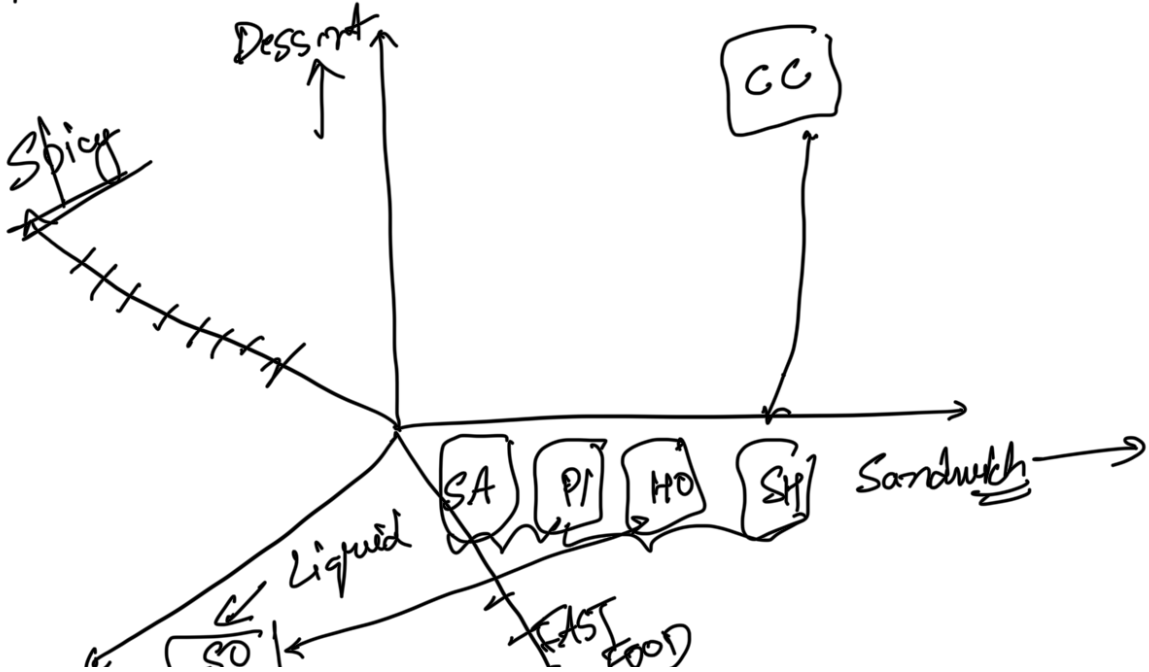
PIZZA SHAWARMA SOUP HOT DOG SALAD $\boxed{1D}$



Chocolate croissant (CC)



Every embedding will represent each item in n-dimensional space using floating point number



Real-world embedding Spaces

$d \rightarrow 2, 3$

$d \rightarrow 256, 512, 1024$

* Embeddings will be specific to the task

FOOD ITEMS

VEG / NON-VEG ITEMS

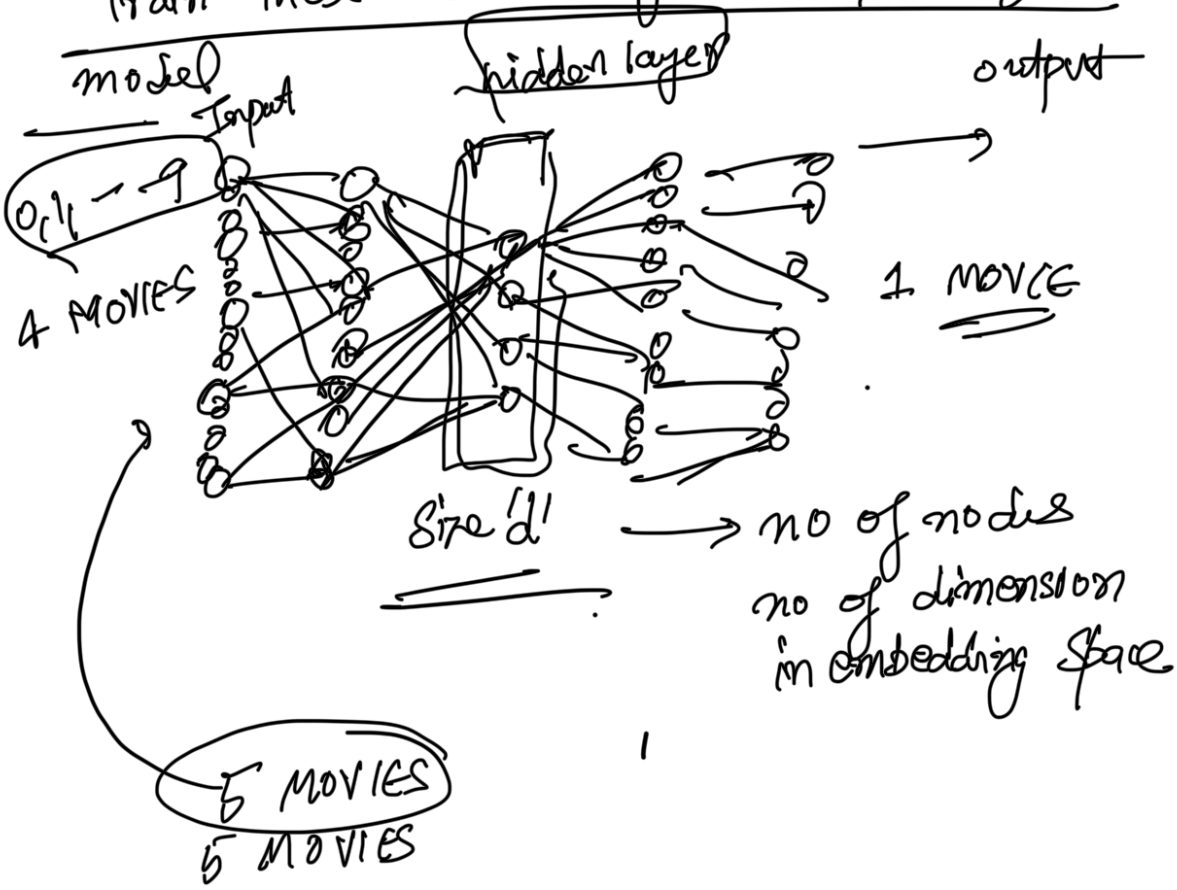
BREAKFAST / LUNCH / DINNER

* Knowledge Distillation

Obtaining Embeddings :

PCA → Principal Component Analysis

Train these embeddings as part of the model



① iii, third and three, 3

② Embedding projector

Orange → fruit
Orange → color

Contextual Embedding:

I want to ____ it orange

I want to ____ the orange

cat / paint